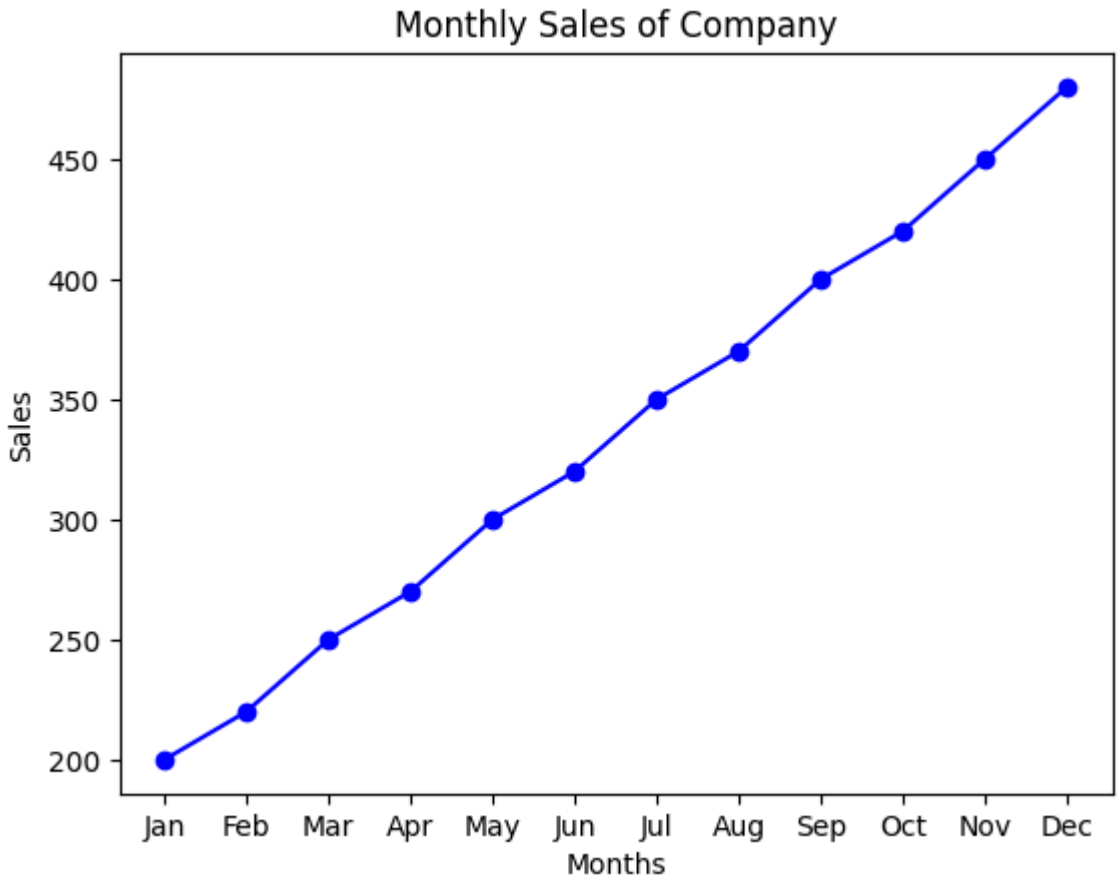
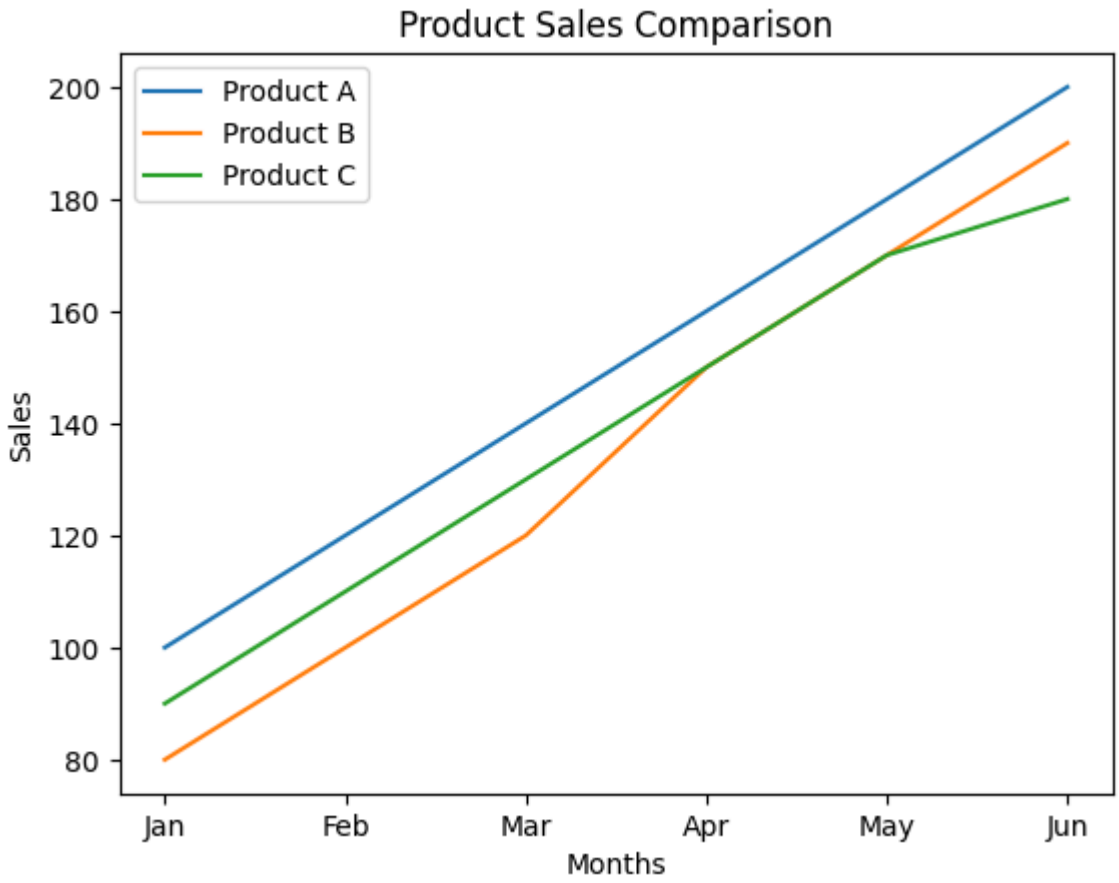
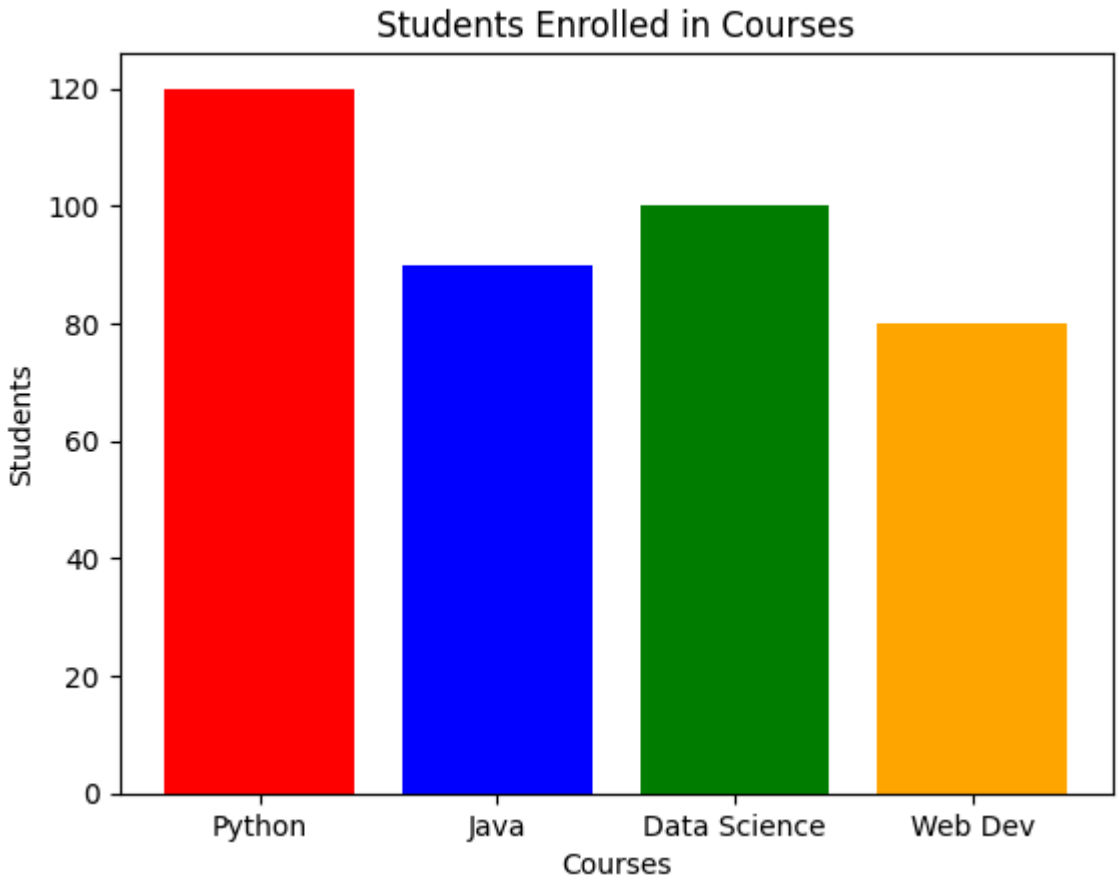
In [1]:
```
#Line Chart - Monthly Sales
import matplotlib.pyplot as plt

months = ["Jan","Feb","Mar","Apr","May","Jun","Jul","Aug","Sep","Oct","Nov","Dec
sales = [200,220,250,270,300,320,350,370,400,420,450,480]

plt.plot(months,sales,marker="o",color="blue")
plt.xlabel("Months")
plt.ylabel("Sales")
plt.title("Monthly Sales of Company")
plt.show()
```



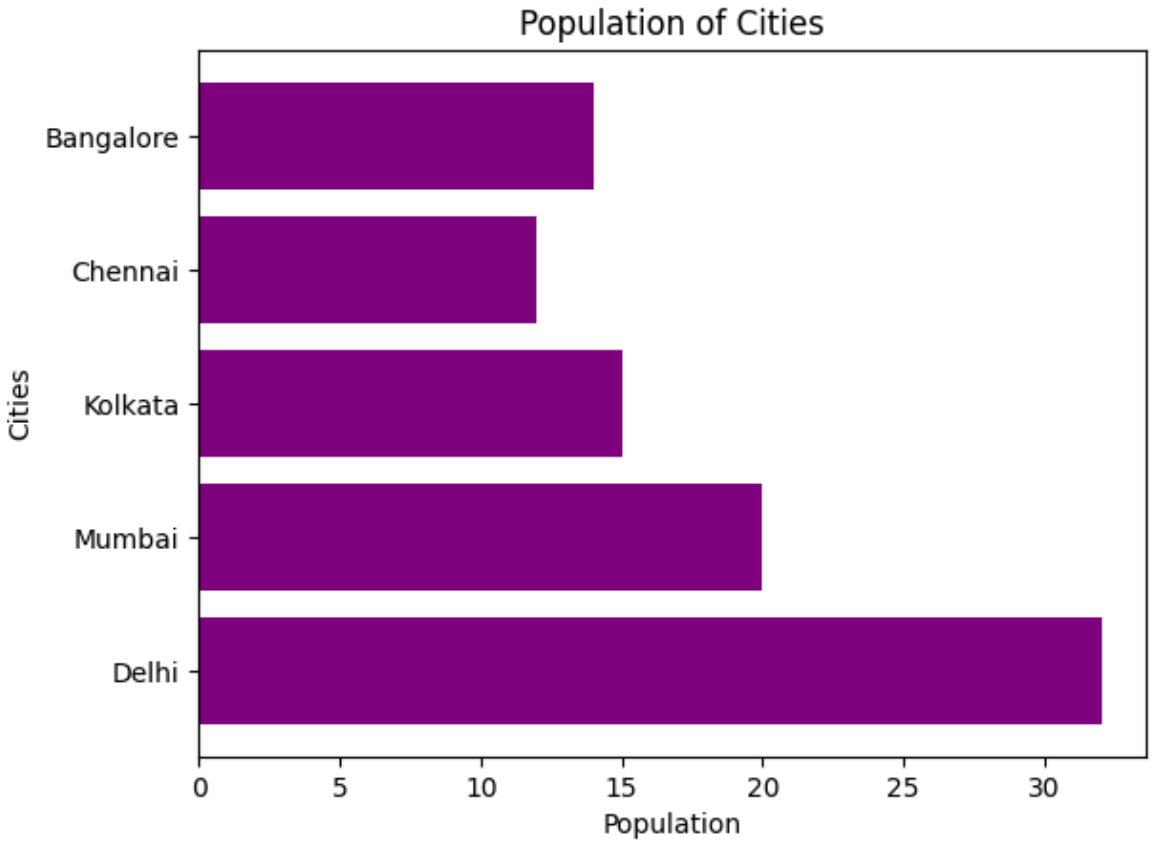
In [2]:
```
#Multiple Line Chart - 3 Products
import matplotlib.pyplot as plt

months=["Jan","Feb","Mar","Apr","May","Jun"]

A=[100,120,140,160,180,200]
B=[80,100,120,150,170,190]
C=[90,110,130,150,170,180]

plt.plot(months,A,label="Product A")
plt.plot(months,B,label="Product B")
plt.plot(months,C,label="Product C")

plt.xlabel("Months")
plt.ylabel("Sales")
plt.title("Product Sales Comparison")
plt.legend()
plt.show()
```



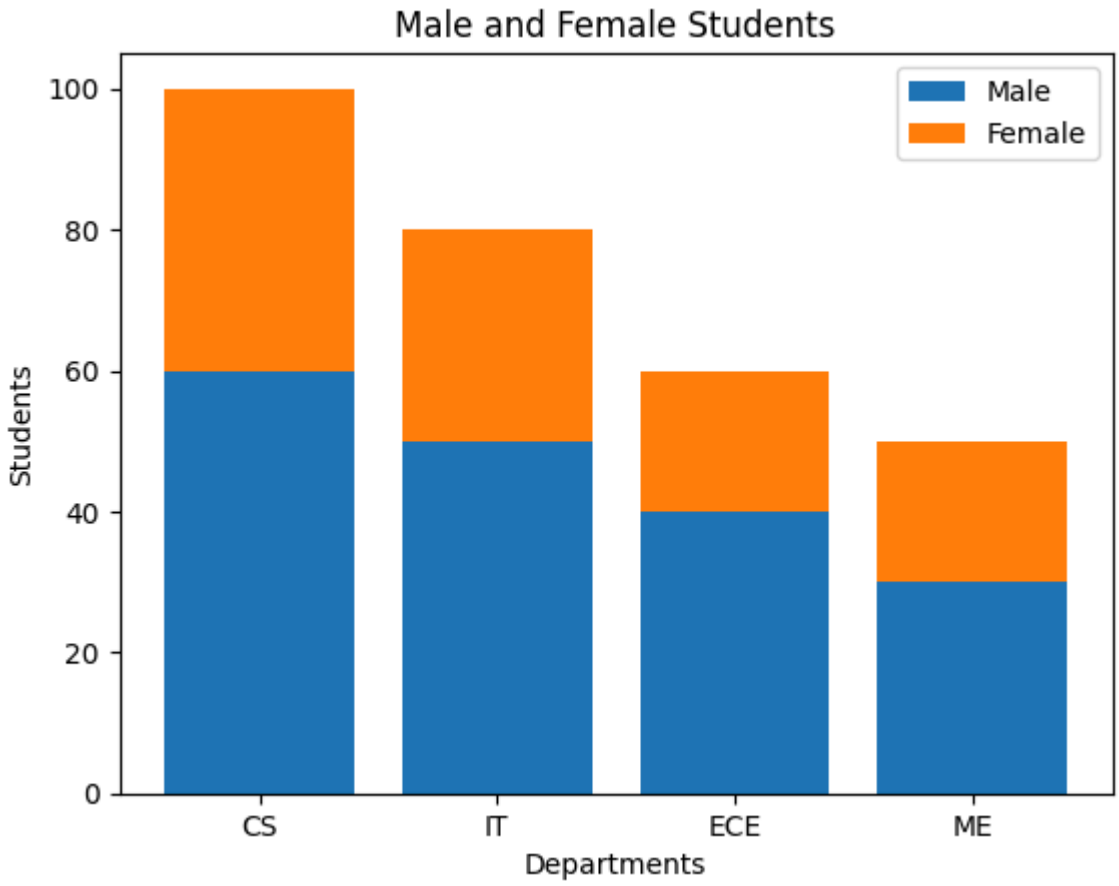
In [3]:
```
#Bar Chart – Students in Courses
import matplotlib.pyplot as plt

courses=["Python","Java","Data Science","Web Dev"]
students=[120,90,100,80]

plt.bar(courses,students,color=["red","blue","green","orange"])
plt.xlabel("Courses")
plt.ylabel("Students")
plt.title("Students Enrolled in Courses")
plt.show()
```



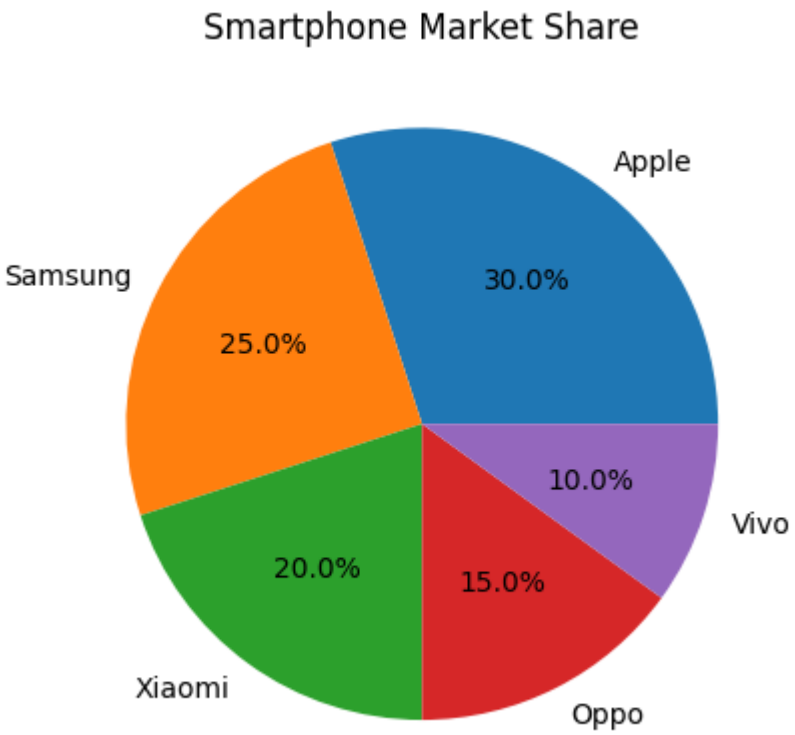
In [5]:
```
#Horizontal Bar Chart
import matplotlib.pyplot as plt

cities=["Delhi","Mumbai","Kolkata","Chennai","Bangalore"]
population=[32,20,15,12,14]

plt.barh(cities,population,color="purple")
plt.xlabel("Population")
plt.ylabel("Cities")
plt.title("Population of Cities")
plt.show()
```



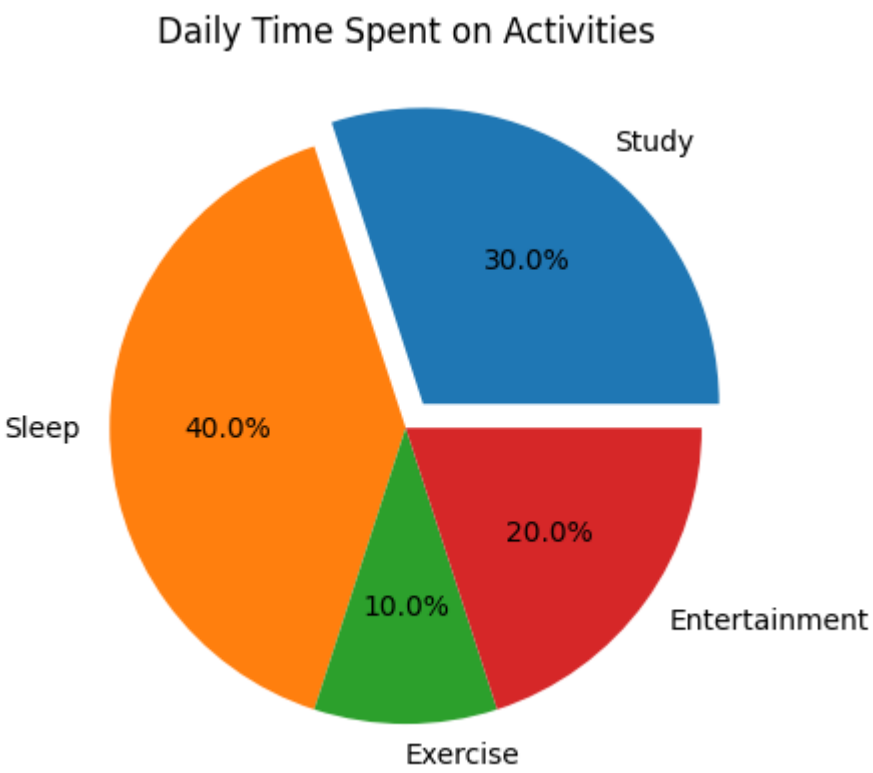
In [6]:
```
#Stacked Bar Chart
import matplotlib.pyplot as plt

dept=["CS","IT","ECE","ME"]
male=[60,50,40,30]
female=[40,30,20,20]

plt.bar(dept,male,label="Male")
plt.bar(dept,female,bottom=male,label="Female")

plt.xlabel("Departments")
plt.ylabel("Students")
plt.title("Male and Female Students")
plt.legend()
plt.show()
```



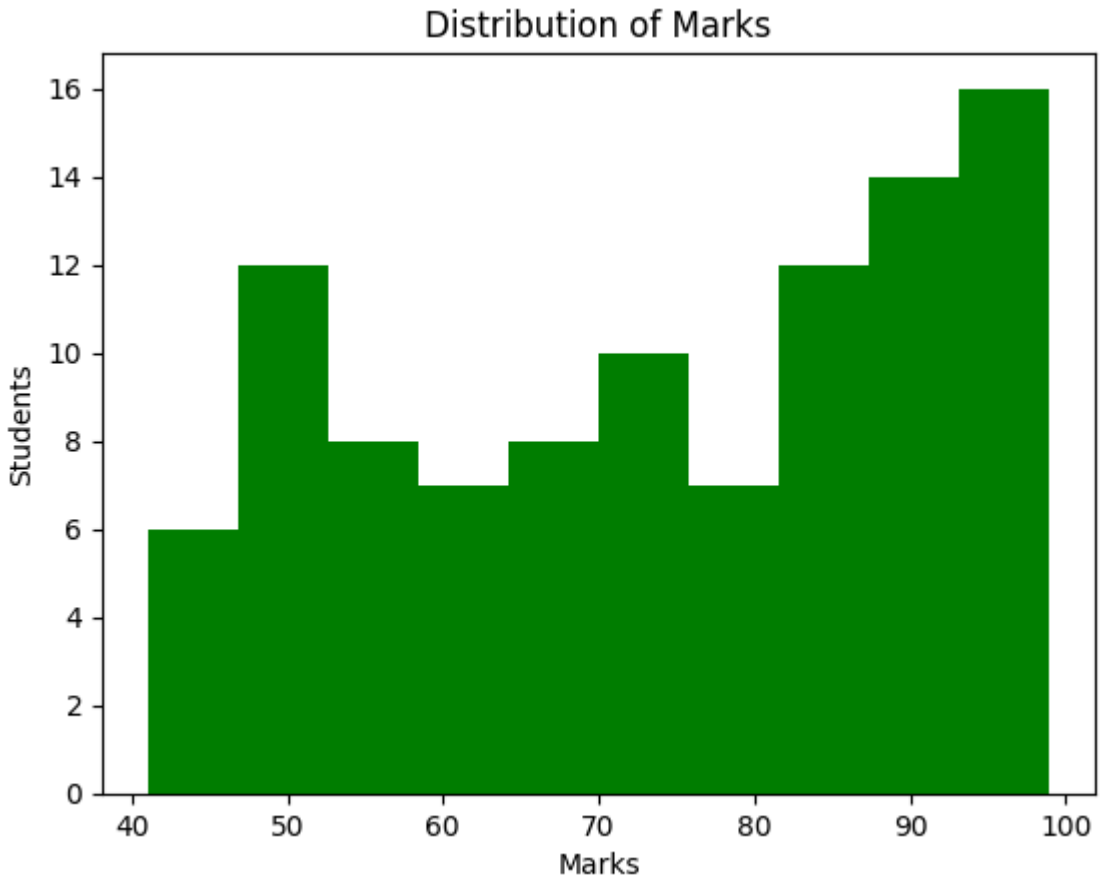
In [7]:
```
#Pie Chart - Smartphone Market Share
import matplotlib.pyplot as plt

brands=["Apple","Samsung","Xiaomi","Oppo","Vivo"]
share=[30,25,20,15,10]

plt.pie(share,labels=brands,autopct="%1.1f%%")
plt.title("Smartphone Market Share")
plt.show()
```



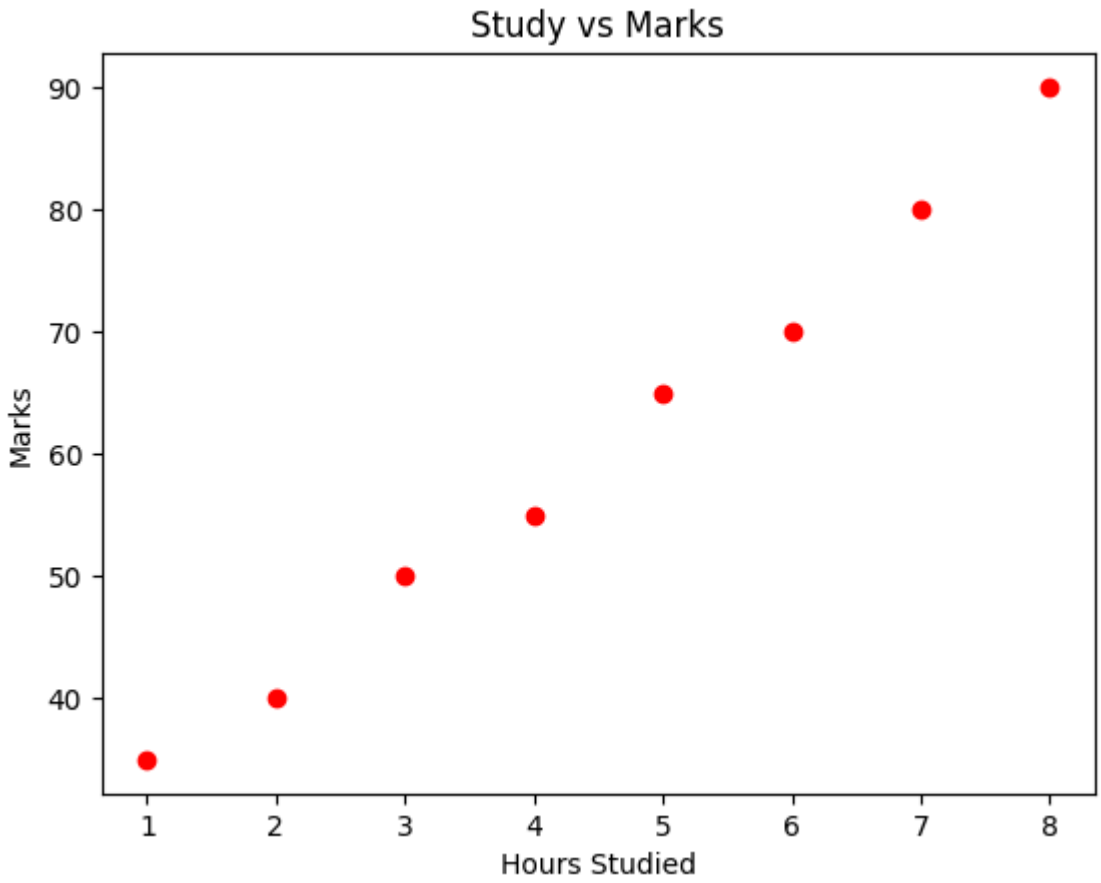
In [8]:
```
#Exploded Pie Chart
import matplotlib.pyplot as plt

activities=["Study","Sleep","Exercise","Entertainment"]
hours=[6,8,2,4]

explode=[0.1,0,0,0]

plt.pie(hours,labels=activities,explode=explode,autopct="%1.1f%%")
plt.title("Daily Time Spent on Activities")
plt.show()
```



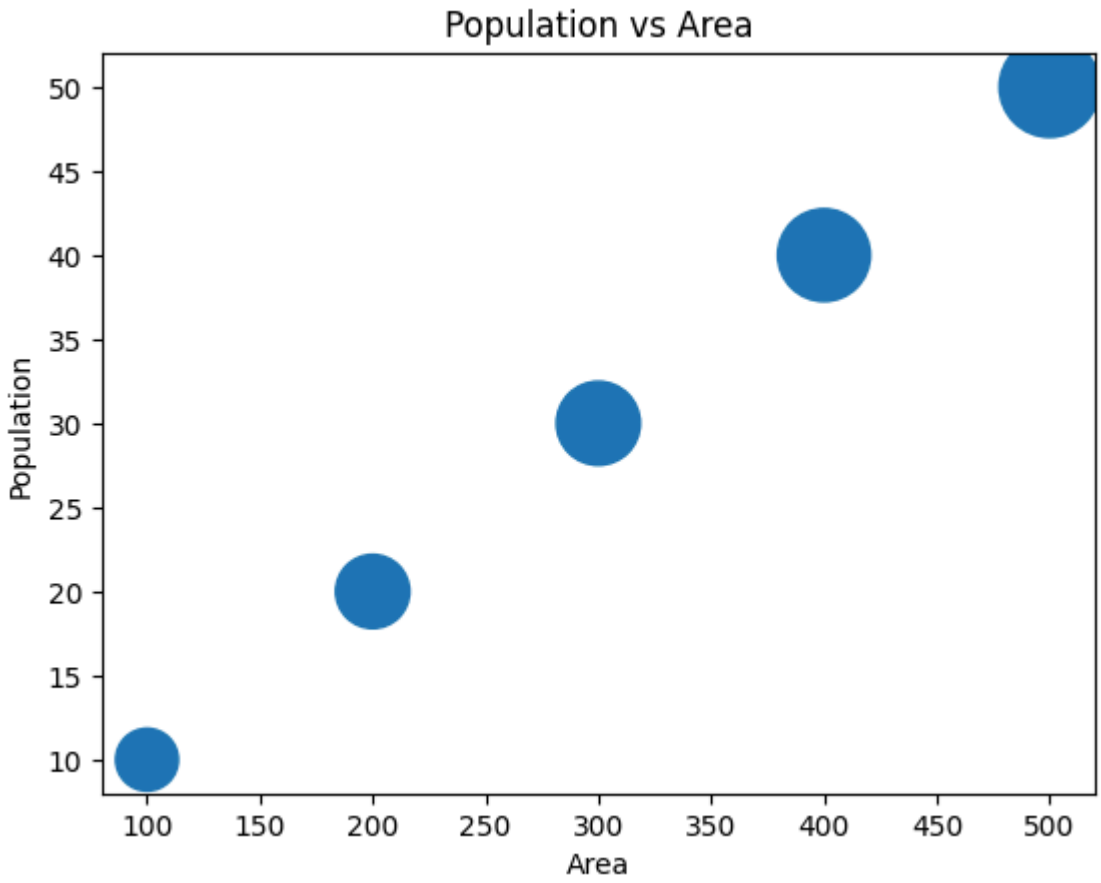
In [9]:
```
#Histogram - Marks Distribution
import matplotlib.pyplot as plt
import numpy as np

marks=np.random.randint(40,100,100)

plt.hist(marks,bins=10,color="green")
plt.xlabel("Marks")
plt.ylabel("Students")
plt.title("Distribution of Marks")
plt.show()
```



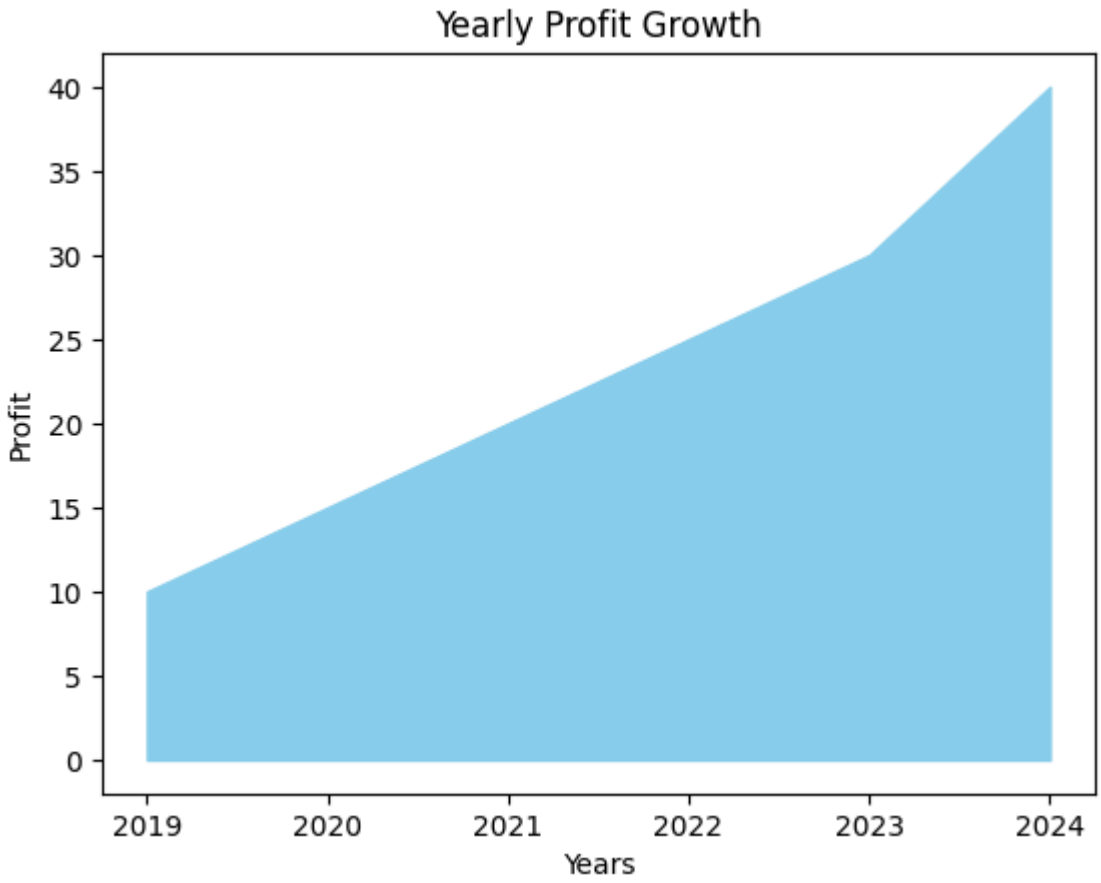
In [10]:
```
#Scatter Plot
import matplotlib.pyplot as plt

hours=[1,2,3,4,5,6,7,8]
marks=[35,40,50,55,65,70,80,90]

plt.scatter(hours,marks,color="red")
plt.xlabel("Hours Studied")
plt.ylabel("Marks")
plt.title("Study vs Marks")
plt.show()
```



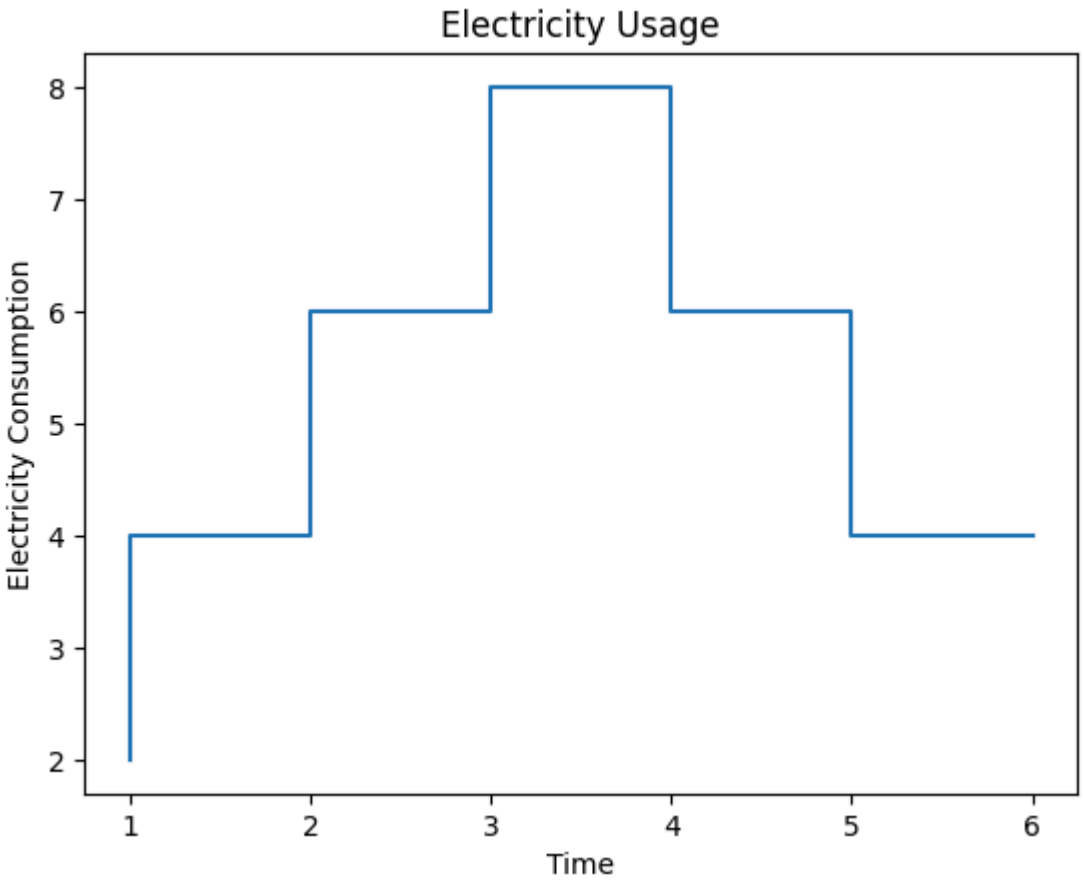
In [11]:
```
#Bubble Chart
import matplotlib.pyplot as plt

cities=["A","B","C","D","E"]
population=[10,20,30,40,50]
area=[100,200,300,400,500]
gdp=[500,700,900,1100,1300]

plt.scatter(area,population,s=gdp)

plt.xlabel("Area")
plt.ylabel("Population")
plt.title("Population vs Area")
plt.show()
```



In [12]:
```
#Area Chart
import matplotlib.pyplot as plt

years=[2019,2020,2021,2022,2023,2024]
profit=[10,15,20,25,30,40]

plt.fill_between(years,profit,color="skyblue")

plt.xlabel("Years")
plt.ylabel("Profit")
plt.title("Yearly Profit Growth")
plt.show()
```



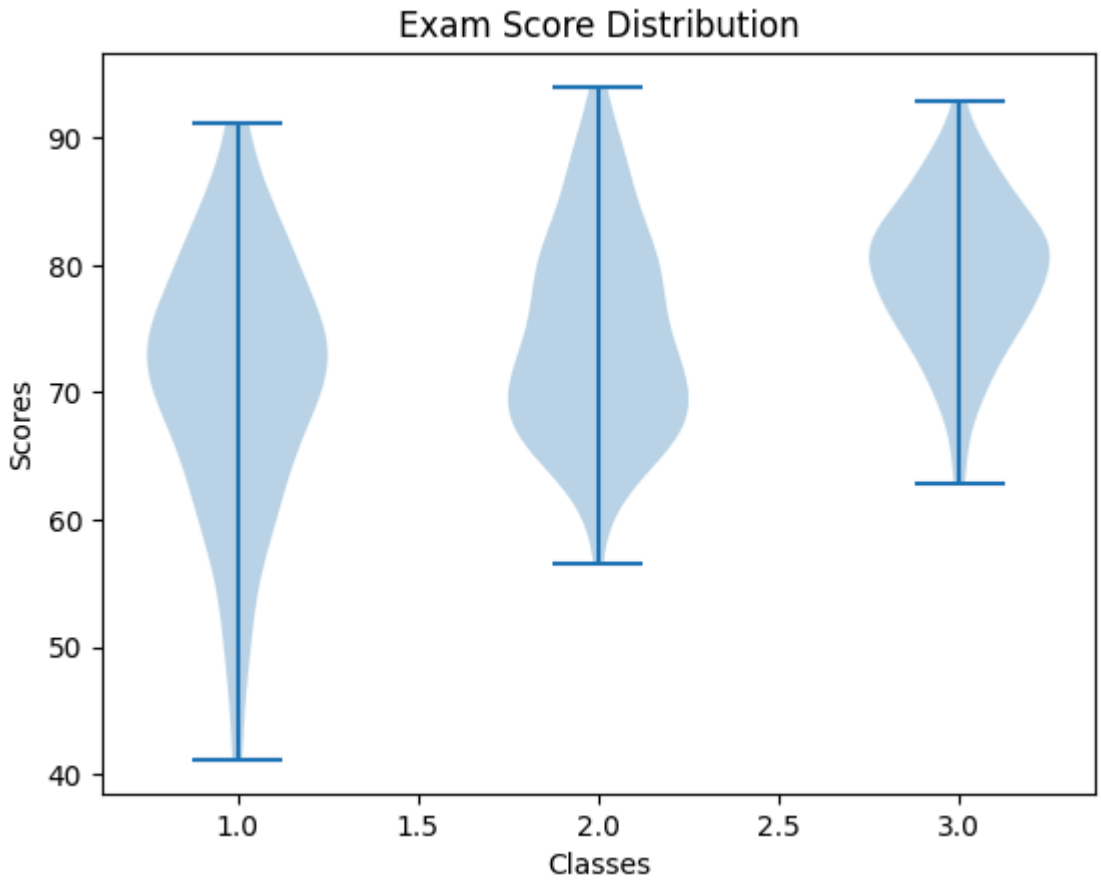
In [13]:
```
#Step Chart
import matplotlib.pyplot as plt

time=[1,2,3,4,5,6]
consumption=[2,4,6,8,6,4]

plt.step(time,consumption)

plt.xlabel("Time")
plt.ylabel("Electricity Consumption")
plt.title("Electricity Usage")
plt.show()
```



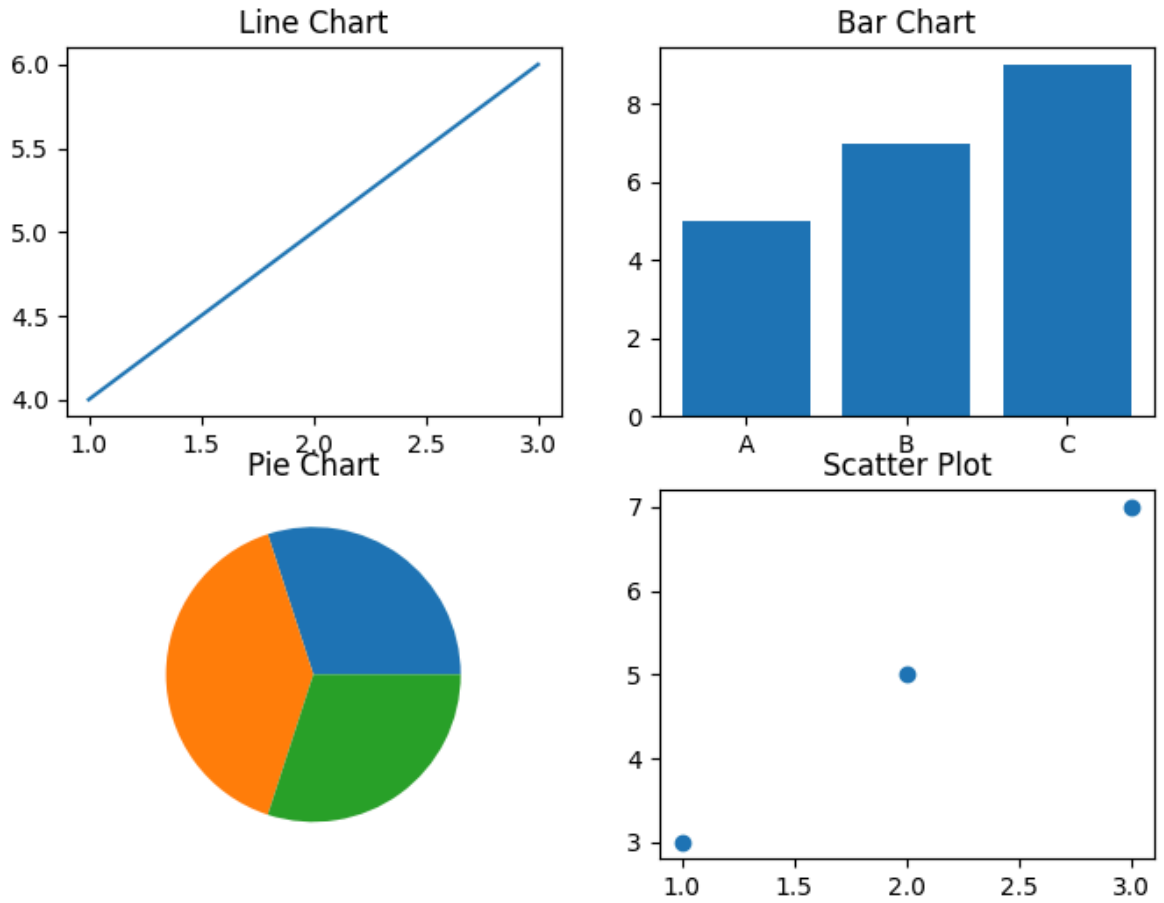
In [14]:
```
#Box Plot
import matplotlib.pyplot as plt

dept1=[25,30,35,40,45]
dept2=[30,35,40,50,55]
dept3=[20,25,30,35,40]

plt.boxplot([dept1,dept2,dept3],labels=["HR","IT","Finance"])

plt.xlabel("Departments")
plt.ylabel("Salary")
plt.title("Salary Distribution")
plt.show()
```

```
C:\Users\ssima\AppData\Local\Temp\ipykernel_19352\3093441869.py:8: MatplotlibDepr
ecationWarning: The 'labels' parameter of boxplot() has been renamed 'tick_label
s' since Matplotlib 3.9; support for the old name will be dropped in 3.11.
  plt.boxplot([dept1,dept2,dept3],labels=["HR","IT","Finance"])
```

In [16]:
```
#Violin Plot
import matplotlib.pyplot as plt
import numpy as np

data=[np.random.normal(70,10,100),
      np.random.normal(75,8,100),
      np.random.normal(80,6,100)]

plt.violinplot(data)

plt.xlabel("Classes")
plt.ylabel("Scores")
plt.title("Exam Score Distribution")
plt.show()
```



In [17]:
```
#Subplots
import matplotlib.pyplot as plt

plt.figure(figsize=(8,6))

plt.subplot(2,2,1)
plt.plot([1,2,3],[4,5,6])
plt.title("Line Chart")

plt.subplot(2,2,2)
plt.bar(["A","B","C"],[5,7,9])
plt.title("Bar Chart")

plt.subplot(2,2,3)
plt.pie([30,40,30])
plt.title("Pie Chart")

plt.subplot(2,2,4)
plt.scatter([1,2,3],[3,5,7])
plt.title("Scatter Plot")

plt.show()
```


Line Chart
6.0
5.5
5.0
4.5
4.0
1.0
1.5
2.0
2.5
3.0
Bar Chart
8
6
4
2
0
A
B
C
Pie Chart
Scatter Plot
7
6
5
4
3
1.0
1.5
2.0
2.5
3.0

In [ ]: